Formalizing AI Scientist Systems: A Unified Framework for Automated Scientific Discovery

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Abstract

The emergence of end-to-end systems capable of automating hypothesis generation, experiment execution, and knowledge synthesis represents a significant change in how science is conducted. Despite rapid empirical progress, AI Scientist systems lack a unifying formal framework, which makes principled comparison across architectures and scientific domains difficult. We propose Servo, a six-component framework (S, G, E, V, M, π) comprising the Search space, hypothesis Generator, Experiment executor, Validator, Memory, and search Policy (π), grounded in the theory of Partially Observable Markov Decision Processes (POMDPs) and Bayesian Experimental Design (BED). We apply this framework to analyze core AI Scientist systems and their instantiations across seven scientific domains (mathematics, physics, chemistry, materials science, biology, social science, and computer science), and separately examine a complementary artifact infrastructure proposal that addresses the output format and verification layer of such systems. Our analysis surfaces three persistent open problems: (1) the non-automatability of novelty evaluation, which remains dependent on human judgment despite multi-layer automated proxies; (2) the absence of BED-optimal search policies, despite the theoretical availability of tractable approximations; and (3) the underdevelopment of semantic memory M_s, whose absence prevents accumulated experimental experience from improving future search. We demonstrate that the completeness of the validator V is the single most consequential design variable: it determines whether closed-loop autonomous operation is feasible in each domain, and current AI Scientist systems can be read as a progression toward more complete validators.

1 Introduction

AI Scientist systems—agents that independently generate hypotheses, execute experiments, and synthesize knowledge—have proliferated rapidly [Boiko et al., 2023, Lu et al., 2024, 2026, Schmidgall et al., 2025, Feng et al., 2026, Liu et al., 2026], yet the field lacks a shared formal vocabulary. Systems range from tool-augmented synthesis agents (Coscientist [Boiko et al., 2023]) through closed-loop manuscript-writing pipelines (The AI Scientist [Lu et al., 2024, 2026]) to formal-mathematics agents (Aletheia [Feng et al., 2026]), each introducing bespoke terminology that makes systematic cross-system comparison difficult. Without a shared decomposition, questions that bear directly on deployment remain unanswered: What determines whether a system can operate in a closed loop? Why does autonomous novelty evaluation fail consistently? What is the theoretically optimal search policy?

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We address these gaps with three contributions: (1) Servo, a formal framework (S, G, E, V, M, π) connecting AI Scientist systems to POMDP and Bayesian Experimental Design theory (Section 4); (2) analysis of four core systems, an artifact infrastructure proposal, and domain-specific instantiations across seven scientific fields (Sections 5, 6); and (3) three open problems, each with a falsification criterion, that constrain progress across all current systems (Section 7). The central finding that emerges from this analysis is that the completeness of the validator V—not the sophistication of the hypothesis generator G or the search policy π—is the primary determinant of closed-loop feasibility.

Scope. We cover systems whose primary goal is automated scientific discovery through hypothesis-experiment-validation cycles, excluding specialized tools within human-conducted pipelines [Jumper et al., 2021].

2 Background

Following Kaelbling et al. [1998], we model AI Scientist systems as partially observable Markov decision processes: the true state s (unknown structure of nature) is inaccessible; the system maintains a belief $b(s)$ updated by experimental observations. Formally, an AI Scientist POMDP is the tuple $(\mathcal{S}, \mathcal{A}, T, R, \Omega, \mathcal{O})$, where $\mathcal{A}$ is the action space (hypothesis generation and experiment execution), T is the transition function, R is the reward (scientific value of discovery), and $\mathcal{O}$ is the noisy observation process. Three properties distinguish this from standard control: R cannot be pre-specified (significance is not defined a priori); $\mathcal{S}$ is unbounded; and T is non-stationary (accumulated knowledge changes the exploration frontier). Bayesian Experimental Design [Chaloner and Verdinelli, 1995, Rainforth et al., 2024] provides the normative criterion for the search policy π: select d^* maximizing Expected Information Gain (EIG),

$$d^* = \arg\max_{d \in \mathcal{D}} \text{EIG}(d) = \arg\max_{d} \mathbb{E}\left[H(\theta) - H(\theta \mid y, d)\right] \quad (1)$$

where $H(\theta)$ is prior entropy over parameters θ and $H(\theta \mid y, d)$ is expected posterior entropy after observing y under d. EIG can be approximated via variational methods [Foster et al., 2019], amortized policies [Foster et al., 2021], and RL [Blau et al., 2022], yet no current AI Scientist system implements it—a gap we identify in Section 7.

3 Related Work

AI Scientist systems. Prior work [Lu et al., 2024, 2026, Schmidgall et al., 2025, Boiko et al., 2023] provides qualitative characterizations of individual systems but no shared framework for cross-system comparison; idea-generation studies [Si et al., 2024, Baek et al., 2024] establish quality baselines without unifying the full discovery loop.

Theoretical foundations. The POMDP framework [Kaelbling et al., 1998] and BED [Chaloner and Verdinelli, 1995, Rainforth et al., 2024] are classical; we are the first to explicitly apply BED as a design target for AI Scientist search policies, connecting POMDP (structural formulation) to BED (normative policy criterion). The reward specification and specification gaming literatures study the consequences of unreliable reward signals; Servo extends this lens to AI Scientist systems by stratifying V into reliability levels and connecting each level to specific closed-loop failure modes.

Formal models and artifact infrastructure. Taniguchi et al. [2025] propose Collective Predictive Coding (CPC-MS) as a social model of science, recasting scientific activity as decentralized Bayesian inference across a community. CPC-MS and Servo are complementary: CPC-MS provides a collective and social model in which peer review is a Metropolis-Hastings consensus process; Servo provides the single-system and modular decomposition that must be instantiated at each node. The three open problems we identify are precisely the necessary conditions for a CPC-MS-compliant collective system to be realizable. Liu et al. [2026] address the artifact layer — output format and verification protocol — rather than the discovery loop itself; we position ARA as a V and M_s infrastructure contribution within Servo, discussed in Sections 5.1 and 7.

4 The Servo Framework

We define an AI Scientist system as a tuple (S, G, E, V, M, π) with the following components:

$$\text{Servo} = (S,\ G,\ E,\ V,\ M,\ \pi) \tag{2}$$

S: Search Space The set of candidate hypotheses, programs, molecules, experimental conditions, or other objects subject to exploration. S may be static or dynamically expanding (open-ended).

$G : M \times S \to h$: Hypothesis Generator A function that proposes new candidate $h \in S$ conditioned on accumulated memory M and the current state of the search space. G may involve LLM generation, evolutionary mutation, or symbolic search.

$E : h \times P \to o$: Experiment Executor A function that executes hypothesis h under protocol P to produce observation o. The fidelity of E ranges from fast computational simulation (low cost, potential surrogate error) to physical wet-lab execution (high cost, ground truth).

$V : o \to r$: Validator A function that assigns a reward signal r to observation o. We distinguish four layers of V:

$$\begin{aligned}
V_{\text{syntax}} &: o \to \{\text{pass}, \text{fail}\} && \text{(deterministic, fast)}\\
V_{\text{semantic}} &: o \to r \in [0,1] && \text{(LLM-based, stochastic)}\\
V_{\text{empirical}} &: o \to r && \text{(reproduction-based)}\\
V_{\text{human}} &: o \to \{\text{novel}, \text{significant}\} && \text{(non-automatizable)}
\end{aligned}$$

M: Memory Structured as (M_e, M_s, M_p). M_e (episodic) stores individual records (h_i, o_i, r_i); M_s (semantic) stores distilled knowledge patterns; M_p (procedural) stores learned protocols. The structure of M directly determines the quality of G and π.

$\pi : b(S) \times M \to h$: Search Policy The decision rule for selecting the next hypothesis. The belief state $b(S)$ is updated after each experiment. In the BED interpretation, $\pi^* = \arg\max_d \text{EIG}(d)$ (Equation 1). In practice, current systems use policies ranging from reactive (ReAct) to greedy to tree search.

The operational loop is:

$$\begin{aligned}
\pi(b, M) \ \to\ G(M_s, S) \ \to\ E(h, P) \ \to\ V(o)&\\
\to\ M \mathrel{+}= (h, o, r) \ \to\ b(S)\ \text{update} \ \to\ \text{repeat}&
\end{aligned} \tag{3}$$

We further introduce $H \in [0, 1]$ (distinct from the entropy function $H(\theta)$ in Eq. 1) as an auxiliary parameter denoting the degree of human intervention: $H = 0$ for fully autonomous systems and $H = 1$ for systems in which humans assume the role of G or π.

V-completeness as a design dimension. The four V layers form a partial order under reliability and automatability. V_{syntax} is fully automatable and deterministic; V_{semantic} is automatable but stochastic and subject to LLM biases; $V_{\text{empirical}}$ varies from well-calibrated (DFT for crystal stability) to poorly calibrated (GNN surrogates for wet-lab outcomes); V_{human} is irreplaceable for novelty and significance but is the rate-limiting step for loop frequency. In domains where correctness is mechanically decidable (formal proof kernels, type-checkers), V_{formal} — a fifth special case — achieves the ideal limit of $V_{\text{empirical}}$: perfect calibration and zero false positives. Table 2 uses V_{formal} for mathematics (formal) to mark this distinct reliability regime. We define V completeness as the highest layer of V that is both accessible and reliably calibrated in a given system and domain. Accessibility without calibration does not count: Agent Laboratory's V_s (LLM reviewer) is accessible but systematically biased, so it contributes imperfect V completeness rather than full V_s completeness. This dimension is the primary design variable because it sets a ceiling on what π can learn and what G can be rewarded for: without a reliable V, no policy can distinguish progress from regression, and no generator can improve with experience. The cross-domain pattern in Section 6 is a direct consequence: every domain's loop status is determined by V completeness, not by the sophistication of its G or π.

5 Analysis of Core AI Scientist Systems

We apply Servo to four end-to-end systems [Boiko et al., 2023, Lu et al., 2024, 2026, Schmidgall et al., 2025] and one artifact infrastructure proposal. Table 1 synthesizes the key dimensions; the cross-system pattern is that every architectural advance is a gain in V completeness, and V completeness is what unlocks closed-loop operation.

	Coscientist	AI Sci. (2024)	AI Sci. (2026)	AgentLab
Novelty measure	None	LLM self-score	3-layer	Human-delegated
Loop closed?	No	Yes (greedy)	Yes (tree)	Partial
π type	ReAct	Greedy	Tree search	Human/seq.
V completeness	V_h only	V_s	$V_e + V_s + V_h$	V_s (biased)
M structure	Ephemeral	M_e	$M_e + M_s$	M_e
H (approx.)	G (≈1)	Low (≈0)	Low (≈0)	G+π (≈0.8)

Table 1: Comparative Servo analysis of four core AI Scientist systems. V_s=semantic, V_e=empirical, V_h=human.

The progression across the four systems is a systematic increase in V completeness, not in G or π sophistication. Coscientist's single GC-MS measurement (V_h only) prevents loop closure entirely. The AI Scientist (Sakana) adds V_s via a 5-ensemble GPT-4o reviewer (balanced accuracy 0.65 vs. human 0.66 [Lu et al., 2024]), enabling the first genuinely closed loop despite greedy π. The AI Scientist (Nature) stacks V_e (replication nodes with mean ± s.d.), V_s (automated reviewer, balanced accuracy 0.69 [Lu et al., 2026]), and actual peer review (V_h via ICLR workshop), achieving the most complete V in the class. Agent Laboratory delegates V to V_s (NeurIPS-guideline LLM reviewer) but documents +2.3-point systematic over-estimation relative to human PhD students—illustrating that adding a V layer does not guarantee V completeness if the layer is unreliable. The three systems that close the loop each do so via a V upgrade; Agent Laboratory, which adds a V_s layer without calibrating it, demonstrates the converse — an accessible but unreliable V does not close the loop. No system in this class achieves improved loop feasibility through G or π advances alone, without a corresponding V improvement.

5.1 Artifact Infrastructure: Agent-Native Research Artifacts

Liu et al. [2026] address a complementary problem: what the output of the AI Scientist loop should look like and how it can be independently verified. Agent-Native Research Artifacts (ARA) replace the narrative paper with a four-layer ontology: `/logic` (claims with falsification criteria), `/src` (executable code), `/trace` (exploration graph: question, decision, experiment, dead_end, pivot), and `/evidence` (raw results). The ARA Seal provides three-level automated verification—V_{syntax} (schema conformance), V_{semantic} (Rigor Auditor, 82.6% average detection rate across five structural violation types), and $V_{\text{empirical}}$ (sandboxed reproduction)—with V_{human} reserved for significance, novelty, and taste: "judgments of significance, novelty, and taste are reserved for human reviewers" [Liu et al., 2026]. The motivation is quantitative: only 45.4% of reproduction requirements are fully specified in published papers, and 90.2% of extension costs are consumed by failed exploration [Liu et al., 2026]. The `/trace` layer addresses M_e and M_s: preserved failure traces accelerated agent performance on extension tasks but also anchored more capable agents to prior solution paths—richer M_e does not monotonically improve π.

6 Domain Analysis

Table 2 maps Servo instantiations across seven scientific domains; V completeness co-varies with loop closability without exception. To illustrate scale: Aletheia [Feng et al., 2026] attempted 700 Erdős problems; of the 200 solutions selected for human audit, 13 (6.5%) were rated meaningfully correct; GNoME [Merchant et al., 2023] screened 2.2 million candidate crystal structures via high-throughput DFT. Systems analyzed span mathematics (formal: [Romera-Paredes et al., 2024, Xin et al., 2024]; NL: [Feng et al., 2026]), physics [Udrescu and Tegmark, 2020, Cranmer et al., 2020], chemistry [Bran et al., 2024, Sprueill et al., 2024], materials [Merchant et al., 2023], biology [O'Donoghue et al., 2023], social science [Manning et al., 2024, Aher et al., 2023, Park et al., 2023], and CS/algorithms [Real et al., 2020, Sartori and Blum, 2024, Jaber and Jaber, 2026].

Domain	V type	V reliability	Loop	π	Primary bottleneck
Math (formal)	V_{formal}	Perfect oracle	Fully closed	RL/MCTS	Pretraining contamination
Math (NL)	$V_{\text{sem}} + V_h$	Imperfect	Closed (low quality)	Seq. refine	Hallucination; plagiarism
Physics	$V_{\text{emp}} + V_h$	Medium	None/partial	Deterministic	No cross-problem M
Chemistry	V_{emp} (surr.)	Medium	Comp. only	Beam	Wet-lab loop open
Materials	V_{emp} (DFT)	High	Comp. closed	Uncert. samp.	DFT wall-time; synth. lag
Biology	$V_{\text{emp}} + V_h$	Low–med	Partial	None	V metrics insufficient
Social Sci.	Stat. test	Low–med	Partial	Factorial	Circular validity
CS / Algo.	V_{emp}	High	Fully closed	Evol./LLM	Interpretability absent

Table 2: Cross-domain Servo analysis across seven domains. V reliability refers to agreement with ground truth.

The domain pattern is not coincidental — it is structurally determined by V type. Mathematics (formal): V_{formal} is binary and unambiguous; RL is stable because reward hacking is impossible by construction, and both FunSearch and DeepSeek-Prover achieve fully closed loops. Mathematics (NL): switching to natural-language proof spaces immediately reintroduces V_{human} as the only reliable oracle; Aletheia's 6.5% meaningful-correctness rate among audited solutions confirms the practical gap. Physics: symbolic interpretability ("this formula is new to cosmologists") is a judgment that no current V can substitute; each problem restarts from scratch because there is no cross-problem M_s. Chemistry and biology: the computational loop closes (GNN surrogate or protocol-accuracy V) but the physical loop does not; NMR, MS, and wet-lab analytical characterization remain human-delegated, and the wet-lab loop is structurally open at the measurement boundary. Materials science: GNoME's DFT oracle is the field's closest approximation to an ideal V—well-calibrated, tractable, and noise-free within its domain; it is not coincidental that this is the single domain where π approximates BED theory. Social science: when G and E share the same underlying model, the evaluator systematically confirms the generator's implicit biases [Aher et al., 2023]; hyper-accuracy distortion means LLM-simulated subjects are too accurate to simulate real human population distributions. CS/algorithms: V_{emp} (runtime speedup, validation accuracy) is high-reliability and enables fully closed loops, but interpretability of discovered algorithms remains absent — knowing that an algorithm works is distinct from understanding why.

7 Open Problems

Our analysis identifies three open problems that persist across all systems and domains.

7.1 The Non-Automatability of Novelty

Novelty cannot be reliably evaluated by automated means: every system from Coscientist (no mechanism) to The AI Scientist Nature (three-layer validation) converges on this conclusion. Two failure modes are documented: LLM self-evaluation bias (systematic overestimation; hallucinated results) and pretraining contamination (systems may "discover" results implicit in training data, as Aletheia documents for Erdős problems). The ARA Seal [Liu et al., 2026] demarcates the automated frontier: its Rigor Auditor achieves 82.6% average detection across five structural violation types, but detects only 22% of orphaned-experiment violations specifically (the two rates measure different denominators and are not complements). It cannot cross the novelty boundary. Concretely: Agent Laboratory's LLM reviewer over-estimated paper quality by +2.3 points on the NeurIPS review scale relative to human PhD students; The AI Scientist (Sakana) hallucinated ablation tables; and of the 200 Erdős solutions selected for human audit (from 700 attempted), only 13 (6.5%) were rated meaningfully correct despite surfacing many superficially plausible candidates.

Falsification criterion: Refuted if an automated V achieves human-level novelty discrimination (inter-rater agreement $\geq$0.80) on a contamination-controlled benchmark spanning $\geq$3 distinct scientific domains.

The novelty boundary resists automation for a deeper reason: novelty, significance, and correctness are three logically distinct properties that current systems conflate. Correctness is verifiable in principle; novelty relative to a knowledge corpus is bounded but contamination-sensitive; significance is irreducibly judgment-dependent. Current systems optimize for correctness proxies while claiming novelty, with the two properties used interchangeably when convenient. Concretely, The AI Scientist [Lu et al., 2024] scores ideas on a single 1–10 "novelty" scale that conflates technical correctness (does the code run?), within-archive novelty (is it different from past ideas?), and scientific significance (is it worth pursuing?)—three logically distinct evaluations collapsed to a single proxy. A viable automated V for novelty requires at minimum: (a) a contamination-controlled reference corpus that separates what the model was trained on from what the field has published; (b) architectural separation between G and V so the evaluating model has not seen its own outputs; and (c) a formal definition of "novel to the field" that does not reduce to "not lexically in training data." None of these conditions hold in any current AI Scientist system.

7.2 The BED–Practice Gap for π

Equation 1 specifies the theoretically optimal policy, yet no current system implements it: all deployed policies are heuristic (ReAct, greedy, tree search, human-guided). The partial exception is GNoME [Merchant et al., 2023], whose deep-ensemble uncertainty sampling approximates EIG over a tractable GNN posterior—but only because DFT provides a well-calibrated V and the hypothesis space admits a learnable probabilistic model, conditions absent from LLM-based systems where the "posterior" is implicit in model weights [Foster et al., 2021]. Concretely, every system in Table 1 degrades to heuristic π: Coscientist selects the next tool reactively, The AI Scientist (Sakana) greedily avoids archive similarity, and Agent Laboratory follows a fixed sequential schedule—none improve with additional experimental data, the failure mode EIG-optimal π would avoid.

Falsification criterion: Refuted if any AI Scientist system implements a policy that explicitly computes and maximizes EIG over a posterior $p(\theta \mid y)$ in a domain where the posterior is not tractable from first principles—ruling out tractable-model proxies such as GNoME's deep-ensemble policy.

The tractability gap has a precise architectural characterization. EIG requires computing $\mathbb{E}_y[H(\theta) - H(\theta \mid y, d)]$, which in turn requires a posterior $p(\theta \mid y)$ over hypothesis parameters θ given experimental outcomes y. In LLM-based systems, θ is implicit in frozen model weights; there is no mechanism to update these weights per experiment, and no tractable approximation to the posterior entropy over the implied hypothesis space. Two paths toward EIG-optimal π exist: (1) maintain an explicit probabilistic surrogate model of the hypothesis space and ground EIG in that model's posterior (as GNoME does for DFT stability predictions), enabling EIG without integrating over model weights; (2) develop amortized EIG estimators [Foster et al., 2021] that bypass explicit posterior computation. The first path is domain-constrained—it requires a learnable, well-calibrated mapping from experimental inputs to outcomes. The second remains an open research problem for open-ended scientific domains where the hypothesis space is not fixed in advance.

7.3 The Underdevelopment of Semantic Memory M_s

Current systems store raw experiment records (M_e) but do not distill them into transferable patterns (M_s): a system with 1,000 experiments does not search more intelligently than one with 10. The ARA `/trace` layer [Liu et al., 2026] is the closest approximation, but traces accelerated extension performance while also anchoring capable agents to prior solution paths—M_s structure must be paired with selective forgetting mechanisms to be beneficial. Concretely, GNoME's GNN—one of the few systems where $G(M_s, S)$ genuinely improves—accumulates domain-specific thermodynamic knowledge after 460K DFT calculations, yet this knowledge does not transfer to chemistry or biology; the system is no more capable in any domain it has not explicitly trained on.

Falsification criterion: Refuted if accumulated cross-experiment knowledge demonstrably improves G-generated hypothesis quality across task instances in a domain where the system was not explicitly pre-trained or purpose-built, measured by a pre-specified metric.

The set-union coverage model—where memory sharing is perfect and additive—is the trivially ideal limit of M_s: knowledge simply accumulates, and the only design question is connectivity. The genuinely hard M_s problems arise when memory is lossy (semantic retrieval degrades with distance from training distribution), compressed (distillation loses evidence the original agent deemed unimportant), conflicting (two agents may reach contradictory conclusions from overlapping experiments), or strategically filtered (incentives to withhold negative results). In each case, accumulated experience can degrade G's performance: a system trained on its own noisy memory may converge prematurely, systematically over-represent successful strategies, or amplify false positives from early exploration. GNoME demonstrates that within-domain M_s accumulation can work: the GNN genuinely improves as DFT outcomes accumulate. The open problem begins where GNoME ends—the same memory structure cannot be reused in chemistry or biology without full retraining, and even within materials science it degrades under distribution shift. Building M_s mechanisms that generalize across experimental contexts, forget selectively, and remain robust to conflicting evidence is a necessary precondition for AI Scientist systems whose $G(M_s, S)$ improves with scale rather than plateauing.

The three open problems are not independent. A system with complete V and tractable M_s would have the necessary prerequisites for an EIG-optimal π: complete V provides the calibrated reward signal that EIG requires, and tractable M_s provides the hypothesis-space model over which EIG is computed. Whether these prerequisites are sufficient for tractable EIG estimation in non-stationary or unbounded S remains open, but they are necessary. They therefore converge on a single structural gap, and progress on any one unlocks disproportionate gains toward the other two. They are also precisely the necessary conditions for a collective AI co-scientist [Taniguchi et al., 2025]: $V_{\text{collective}}$ requires solved novelty automatability; a principled collective π requires closed BED–practice gap; and the shared semantic memory w requires developed M_s. ARA [Liu et al., 2026] may be the most tractable entry point, simultaneously externalizing M_s and calibrating the sub-novelty layers of V.

8 Discussion

Design hierarchy. Our analysis establishes a clear priority ordering: V design precedes all other decisions, because it determines whether closed-loop operation is feasible and what form π can take. In domains where V is noisy or fallible (chemistry, biology, social science), even EIG-optimal π and perfectly structured M_s cannot close the loop—the reward signal is too unreliable to sustain RL or guide principled search. Conversely, in mathematics (formal), V_{formal} is so well-specified that even greedy G and simple π achieve closed loops. The practical implication: before optimizing G or π, designers should ask what level of V completeness the domain actually supports.

Human oversight. Rather than treating $H \in [0, 1]$ as a limitation to be minimized, our analysis suggests that explicit human integration at the V_{human} and π levels is both practically necessary and theoretically appropriate. Human judgment provides access to an information source that no automated proxy currently replicates—particularly for the non-automatizable components of novelty and significance. Systems that design H explicitly (as ARA does by reserving significance and novelty for human reviewers) may achieve better long-run outcomes than those that attempt to minimize it. The goal is not full autonomy but appropriate autonomy: closing the loop at the layers where V is reliable while preserving human judgment at the layers where it is not.

Towards collective AI co-scientists. All systems analyzed operate as single-agent loops. A natural extension is a collective AI scientist: a multi-agent system $\{\text{Servo}_k\}_{k=1}^{K}$ sharing an external semantic memory w and coordinating through a collective validator $V_{\text{collective}}$ [Taniguchi et al., 2025]. The three open problems we identify are precisely the necessary conditions for such a system: $V_{\text{collective}}$ requires solved novelty automatability; a principled collective π requires closing the BED–practice gap; and the shared w requires developed M_s. ARA [Liu et al., 2026] may be the most tractable entry point as a structured artifact protocol for w, simultaneously externalizing M_s and calibrating the sub-novelty layers of V.

Limitations. Servo abstracts away several important factors. The cost structure of E varies by orders of magnitude across domains (milliseconds for code execution, hours for DFT, weeks for wet-lab) and is not modeled. The non-stationarity of S (open-ended problems) is treated as a property to be noted but not formalized. The single-agent framing abstracts away the social layer that Taniguchi et al. [2025] formalize; extending Servo to multi-agent settings is a natural direction for future work. Finally, the cross-domain comparison relies on publicly reported system descriptions; component characterizations may not capture all implementation details.

9 Conclusion

We introduced Servo (S, G, E, V, M, π), a formal framework connecting AI Scientist systems to POMDP and BED theory, and applied it to four core systems, one artifact infrastructure proposal, and seven scientific domains. The central finding is that V completeness is the primary determinant of closed-loop feasibility: every architectural advance across surveyed systems—from Coscientist's single-measurement V_h to The AI Scientist Nature's three-layer validation—is ultimately a gain in V completeness, not in G sophistication or π optimality.

The Servo decomposition provides a shared vocabulary that makes previously implicit cross-system comparisons explicit. Under this vocabulary: the reason mathematics (formal) has the most mature AI Scientist deployments is V_{formal}'s binary reliability, not domain simplicity; the reason chemistry and biology remain loop-open despite capable G and π implementations is V's dependence on physical measurement; and the reason no system improves intelligently with scale beyond its specific training domain is the absence of transferable M_s mechanisms (GNoME's within-domain improvement is the exception that proves the rule: it works only because 460K DFT calculations were performed in a single, well-defined chemical space). These are structural constraints, not engineering gaps to be solved by better language models alone.

The three open problems—novelty non-automatability, the BED–practice gap, and M_s underdevelopment—are not independent, and they are not equally tractable. ARA's Rigor Auditor represents meaningful progress on the sub-novelty layers of V_{semantic}; GNoME's uncertainty sampling demonstrates that EIG-adjacent π is achievable when a tractable probabilistic model of the hypothesis space exists. These partial exceptions suggest that the bottleneck is not fundamental impossibility but missing architectural components—components that Servo now makes precisely nameable. The organizing question for the field remains: how do we build validators and memory structures that are simultaneously complete enough to support closed-loop operation and structured enough to support principled search?

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Conference For AI Scientists 2026 - AI Involvement Checklist

This checklist documents the role of AI systems at each stage of this research. The checklist has two parts: a Research Stage Assessment (items 1–4) rating AI involvement and iteration effort, and an AI System Documentation section (items 5–6).

Research Stage Assessment

1. Hypothesis development.

   Answer: [B]

   Iteration Effort: [Medium]

   Explanation: The core Servo framework tuple (S, G, E, V, M, π), the V-completeness thesis, the H parameter, and the three open problems were conceived by the human author. A frontier LLM agent elaborated framework components, identified literature-level instantiations, and proposed theoretical connections. The human author directed all conceptual decisions and made several substantive corrections to the AI's elaborations.

2. Experimental design and implementation.

   Answer: [B]

   Iteration Effort: [Medium]

   Explanation: This submission contains no computational experiments. The methodology for cross-system analysis (system selection criteria, component extraction protocol, domain comparison structure) was designed by the human author. A frontier LLM agent executed literature retrieval, extracted framework components from primary sources, and performed citation-level fact verification across four verification sweeps.

3. Analysis of data and interpretation of results.

   Answer: [C]

   Iteration Effort: [Medium]

   Explanation: The LLM agent extracted Servo components for all analyzed systems from primary sources and performed citation-level fact verification (4 verification sweeps). The human author reviewed all extractions, corrected mischaracterizations, and synthesized the cross-domain pattern. The V-completeness thesis emerged from the human author's synthesis.

4. Writing.

   Answer: [C]

   Iteration Effort: [High]

   Explanation: The LLM agent drafted the majority of the manuscript text through multiple revision cycles; the human author provided key conceptual passages (framework definitions in §4, core open problems in §7), directed structural decisions, reviewed all drafts, and made final editorial judgments.

AI System Documentation

5. AI systems used.

   Description: A frontier large language model with code execution and long-document processing capabilities was used as the primary research and writing assistant. The system operated as an interactive CLI agent with file-system access, able to read primary sources, execute and debug Python simulations, and iteratively revise LaTeX manuscripts.

6. Observed AI Limitations.

   Description: (1) Systematic citation fabrication: the agent generated plausible but non-existent quotes and statistics, requiring four dedicated citation-verification sweeps using parallel sub-agents. (2) Analytical overconfidence: the agent initially overclaimed empirical significance for mechanically-entailed theoretical consequences; multiple adversarial review rounds were needed to correct framing and remove overclaims. (3) Context drift: over long sessions, the agent occasionally regressed previously corrected errors, requiring explicit re-verification of integrity items before each commit.

Conference For AI Scientists 2026 - Reproducibility and Responsibility Checklist

1. Claims

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction explicitly state the three contributions (framework, cross-domain analysis, open problems). The V-completeness thesis is presented as a cross-domain empirical pattern supported by natural comparison, not a proven theorem, and the paper explicitly acknowledges this scope in §8.

2. Limitations

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Limitations are discussed in §8 (framework abstracts domain cost structures, social layer, non-stationarity) and throughout the domain analysis (§6).

3. Theory assumptions and proofs

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete proof?

   Answer: [NA]

   Justification: The paper makes no formal theorem claims. The V-completeness thesis is a cross-domain empirical pattern supported by natural comparison, not a proven theorem.

4. Experimental result reproducibility

   Question: Does the paper fully disclose all information needed to reproduce the main experimental results?

   Answer: [Yes]

   Justification: This paper contains no computational experiments. The cross-domain analysis methodology (component extraction from primary sources) is fully described in §5 and §6; all analyzed papers are publicly accessible.

5. Open access to data and code

   Question: Does the paper provide open access to data and code?

   Answer: [No]

   Justification: This submission contains no computational experiments. All analyzed papers are publicly accessible, and the framework can be applied following the component definitions in §4.

6. Experimental setting/details

   Question: Does the paper specify all details necessary to understand the results?

   Answer: [Yes]

   Justification: The cross-domain analysis methodology is described in §5 and §6. No computational experiments were performed in this submission.

7. Experiment statistical significance

   Question: Does the paper report error bars or statistical significance information?

   Answer: [NA]

   Justification: This paper contains no computational experiments and reports no statistical results requiring error bars or significance tests.

8. Experiments compute resources

Question: Does the paper provide sufficient information on compute resources?

Answer: [NA]

Justification: This submission contains no computational experiments. The meta-analysis requires only standard document processing and no specialized compute.

9. Research Ethics

   Question: Does the research conform with the NeurIPS Code of Ethics, except where CAISc's AI authorship policies apply?

   Answer: [Yes]

   Justification: This is a meta-analysis of published systems with no human subjects, no proprietary data, and no safety risks. AI authorship is documented per CAISc policy.

10. Broader impacts

    Question: Does the paper discuss both potential positive and negative societal impacts?

    Answer: [Yes]

    Justification: §8 addresses both the potential for accelerating scientific discovery and the risk of premature convergence under widespread AI Scientist deployment (particularly via the M_s underdevelopment problem and novelty non-automatability).